

# Nophenin

Paracetamol Tablet 500 mg

## NAME AND STRENGTH OF ACTIVE SUBSTANCE

Each tablet contains Paracetamol 500 mg.

## PRODUCT DESCRIPTION

White, round flat face bevel edge, 12 mm in diameter tablet, plain on both sides.

## PHARMACODYNAMICS

Pharmacotherapeutic group:  
Other analgesics and antipyretics, anilides  
ATC code: N02B E01

### Mechanism of action

Paracetamol is a type of analgesic and antipyretic. Its mechanism of action is believed to include inhibition of prostaglandin synthesis, particularly in the central nervous system.

### Pharmacodynamic effects

Lack of peripheral prostaglandin inhibition results in important pharmacological properties such as preservation of protective prostaglandins in the gastrointestinal tract.

Therefore, paracetamol is particularly suitable for patients with a history of illness or who are taking concomitant medications, where peripheral prostaglandin inhibition is not required (such as, for example, those with a history of gastrointestinal bleeding or the elderly).

## PHARMACOKINETICS

### Absorption

Paracetamol is readily absorbed from the gastrointestinal tract.

### Distribution

Peak plasma concentrations occur about 10 to 60 minutes after oral doses. Paracetamol is distributed into most body tissues. It crosses the placenta and is present in breast milk. Plasma-protein binding is negligible at usual therapeutic concentrations but increases with increasing concentrations.

### Metabolism

It is metabolised in the liver. A minor hydroxylated metabolite which is usually produced in very small amounts by mixed function oxidases in the liver and which is usually detoxified by conjugation with liver glutathione may accumulate following paracetamol overdosage and cause tissue damage.

### Elimination

It is excreted in the urine, mainly as the glucuronide and sulphate conjugates. The elimination half-life varies from about 1 to 4 hours.

## INDICATION

For pain relief from mild to moderate including; headache, migraine, back pain, musculoskeletal pain, menstrual pain, osteoarthritis pain, toothache, pain after dental procedures/ tooth extraction, pain after vaccination and discomfort due to colds and sore throats. To relieve fever.

## RECOMMENDED DOSAGE

Adults (including the elderly) and children aged 12 years and over:

1-2 tablets (500mg-1000mg paracetamol), every 4-6 hours when necessary.

Children under 12 years of age:  
Not recommended for children under 12 years of age.

- Do not exceed the recommended dose.
- The lowest effective dose and the shortest duration of treatment should be used.
- Interval between doses minimum: 4 hours.
- Maximum daily dose: 4000mg paracetamol (8 tablets).

### Renal impairment:

Patients diagnosed with kidney problems must seek medical advice before taking this medicine. Limitations related to the use of paracetamol products in patients with kidney problems are a result of the content of paracetamol in the medicine.

### Liver impairment:

Patients diagnosed with liver problems must seek medical advice before taking this medicine. Limitations related to the use of paracetamol products in patients with liver problems are a result of the content of paracetamol in the medicine.

## ROUTE OF ADMINISTRATION

Oral

## CONTRAINDICATIONS

- Patient with hypersensitivity to paracetamol or any of the/ other ingredients/ components of the product.
- Patient with severe and active hepatic impairment.

## WARNING AND PRECAUTIONS

This preparation contains PARACETAMOL. Do not take any other PARACETAMOL containing medicines at the same time.

### Allergy alert:

Paracetamol may cause severe skin reactions. Symptoms may include skin reddening, blisters, or rashes. These could be signs of serious conditions.

If these reactions occur, stop use and seek medical assistance right away.

- Keep out of reach of children.
- Do not take if allergic to paracetamol.
- Patients should contact their health care provider if symptoms persist (if the pain lasts for more than 10 days, if there is redness or fever lasts more than 3 days).
- Paracetamol should be given with care to patients with impaired kidney or liver function.
- Large doses should be avoided in patients with hepatic impairment. Paracetamol overdose may harm the liver.
- Do not exceed recommended dose.
- It should be given with care to patients with alcohol dependence.
- Paracetamol provides symptomatic relief only, additional therapy to treat the cause of the pain or fever should be initiated when necessary.
- Patients should be advised to consult their doctor if their headaches become persistent.
- Patients should be advised to consult a doctor if they suffer from non-serious arthritis and need to take painkillers every day.
- Caution should be exercised in patients with glutathione depleted states, as the use of paracetamol may increase the risk of metabolic acidosis.

## INTERACTIONS WITH OTHER MEDICAMENTS

- Anticoagulants – the effect of warfarin and other coumarins may be enhanced by prolonged regular use of paracetamol with increased risk of bleeding. Occasional doses have no significant effect.
- Metoclopramide – may increase speed of absorption of paracetamol.
- Domperidone – may increase speed of absorption of paracetamol.
- Colestyramine – may reduce absorption if given within one hour of paracetamol.
- Imatinib – restriction or avoidance of concomitant regular paracetamol use should be taken with imatinib.
- Flucloxacillin – as concurrent intake has been associated with high anion gap metabolic acidosis (HAGMA), especially in patients with severe renal impairment, sepsis, malnutrition, and other sources of glutathione deficiency (e.g. chronic alcoholism), as well as those using maximum daily doses of paracetamol.

## PREGNANCY AND LACTATION

### Pregnancy

As with the use of other medicines during pregnancy, pregnant women should seek medical advice before taking paracetamol. The lowest effective dose and the shortest duration of treatment should be considered.

### Lactation

Paracetamol is excreted in breast milk but not in a clinically significant amount. Published data do not preclude breastfeeding.

## SIDE EFFECTS

Stop taking this medicine and tell your doctor right away if you:

- experiencing allergic reactions such as skin rash or itching, sometimes accompanied by breathing problems or swelling of the lips, tongue, throat or face
- have a skin rash or peeling or ulcers in the mouth
- have had respiratory problems with aspirin or non-steroidal anti-inflammatory drugs, and experienced a similar reaction to this product
- have bruises or bleeding of unknown cause.

These reactions are rare.

From post-marketing data, the following reactions are very rare:

| Body system                                      | Adverse reactions                                                                                                                                             | Frequency |
|--------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| Blood and lymphatic system disorders             | Thrombocytopenia<br>agranulocytosis                                                                                                                           | Very rare |
| Immune system disorders                          | Anaphylaxis, Cutaneous hypersensitivity reaction including skin rashes and angioedema, Stevens-Johnson syndrome/ toxic epidermal necrolysis has been reported | Very rare |
| Respiratory, thoracic, and mediastinal disorders | Bronchospasm*                                                                                                                                                 | Very rare |
| Hepatobiliary disorders                          | Hepatic dysfunction                                                                                                                                           | Very rare |

\* There have been cases of bronchospasm with paracetamol, but these are more likely in asthmatics sensitive to aspirin or other NSAIDs.

#### SYMPTOMS AND TREATMENT OF OVERDOSE

Liver damage is possible in adults who have taken 10g or more of paracetamol, ingestion of 5g or more of paracetamol may lead to liver damage if the patient has risk factors (see below).

##### Risk factors

If the patient:

- Is on long term treatment with carbamazepine, phenobarbitone, phenytoin, primidone, rifampicin, St John's Wort, or other drugs that induce liver enzymes; or
- Regularly consumes ethanol in the excess of recommended amounts; or
- Is likely to be glutathione deplete e.g. eating disorders, cystic fibrosis, HIV infection, starvation, cachexia.

##### **Symptoms:**

Symptoms of paracetamol overdosage in the first 24 hours are pallor, nausea, vomiting, anorexia, and abdominal pain.

Liver damage may become apparent 12 to 48 hours after ingestion. Abnormalities of glucose metabolism and metabolic acidosis may occur. In severe poisoning, hepatic failure may progress to encephalopathy, disseminated intravascular coagulation, haemorrhage, hypoglycaemia, cerebral oedema, gastrointestinal bleeding and death. Acute renal failure with acute tubular necrosis, strongly suggested by loin pain, haematuria and proteinuria may develop even in the absence of severe liver damage.

Cardiac arrhythmia and pancreatitis have been reported.

##### **Treatment:**

Immediate treatment is essential in the management of paracetamol overdose. Despite a lack of significant early symptoms, patient, should be referred to hospital urgently for immediate medical attention. Symptoms may be limited to nausea or vomiting and may not reflect the severity of overdose or the risk of organ damage. Management should be accordance with established treatment guideline.

Treatment with activated charcoal should be considered if the overdose has been taken within 1 hour. Plasma paracetamol concentration should be measured at 4 hours or later after ingestion (earlier concentrations are unreliable).

Treatment with N-acetylcysteine may be used up to 24 hours after ingestion of paracetamol, however, the maximum protective effect is obtained up to 8 hours post-ingestion. The effectiveness of the antidote declines sharply after this time. If required the patient should be given intravenous N-acetylcysteine, in line with the established dosage schedule.

If vomiting is not a problem, oral methionine may be a suitable alternative for remote areas, outside hospital. Management of patients who present with serious hepatic dysfunction beyond 24 hours from ingestion should be discussed further with the non-pharmaceutical interventions or a liver unit.

Further measures will depend on the severity, nature and course of clinical symptoms of paracetamol intoxication and should follow standard intensive care protocols.

#### EFFECTS ON ABILITY TO DRIVE AND USE MACHINE

Less likely to affect the ability to drive and use machines.

#### STORAGE CONDITIONS

Store below 30°C. Protect from light.

#### SHELF LIFE

Product should not be used beyond the expiry date imprinted on the product packaging.

#### DOSAGE FORMS AND PACKAGING AVAILABLE

Blister pack of 100 x 10's tablets.

#### PRODUCT REGISTRATION HOLDER / MANUFACTURER



##### **GOODSCIENCE SDN BHD**

No. 7, Jalan KPK 4/3, Kawasan Perindustrian Kundang, Kundang Jaya, 48020 Rawang, Selangor, Malaysia.

#### DATE OF REVISION

24<sup>th</sup> March 2025

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